

Systematic Pair-Level Campaign for δ_{pair} : Convergence, Inter-Pair Concentration, and Normalization Structure

O25 — Spectral Admissibility Sub-programme

Jérôme Beau

Independent Researcher

jerome.beau@cosmochrony.org

2026

Abstract

The O-series establishes the transfer chain $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$ unconditionally with respect to the fibre structure of the non-injective projection Π [13]. The value $\delta_{\text{pair}} \approx 7.44$ extracted in O16 [4] was based on a single conjugate pair per prime and a limited prime range. The present paper reports the first systematic campaign computing δ_{pair} across *all* $(q-1)/2$ conjugate pairs $(c, q-c)$ for $q \in \{29, 61, 101, 151\}$, with $M = 50$ independent block samples per pair. Three results are established. First, $\delta_{\text{pair}}(q)$ converges robustly: the inter-pair standard deviation decreases from 0.54 at $q = 29$ to 0.14 at $q = 151$, confirming that δ_{pair} is a structural invariant of the Weil representation rather than a block-level fluctuation. Second, empirical fits of the form $\delta_\infty + a/\log q$ describe the data excellently over the tested range, but carry no asymptotic meaning: the leading finite-size correction is governed by the BFS window depth $n_1(q)$, which satisfies $n_1(q) = \Theta(q)$ by the Window-Depth Theorem [8], making the correction approximately constant in $\log q$ and rendering direct extrapolation structurally degenerate. Third, when the O14 normalization correction $\eta \log q / \log n_1(q)$ [7] is applied, the corrected values $\delta_{\text{corr}}(q) \in [7.7, 9.0]$ fall within the admissible window $[7.4, 10.6]$ consistently across $q \in \{61, 101, 151\}$. This supports the interpretation that the apparent decrease of raw $\delta_{\text{pair}}(q)$ is a finite-size normalization effect rather than a signal of asymptotic drift below the admissible window. The identification of the asymptotic value δ_∞ requires analytical control of $n_1(q)/q$, identified here as the central open direction.

Keywords. Cosmochrony; spectral admissibility; pair-level observable; Weil representation; Heisenberg graphs; capacity exponent; convergence; inter-pair concentration; normalization correction; window depth; BFS; asymptotic analysis

Contents

1	Position of the Problem	2
1.1	What O16–O24 established	2
1.2	Scope of O25	3
2	Setup and Protocol	3
2.1	Observable and block parameterisation	3
2.2	Pair protocol	3
2.3	Fitting window and BFS parameters	4
3	Numerical Results	4
3.1	Main results table	4
3.2	Inter-pair concentration	4
3.3	q -dependence of $\bar{\delta}_{\text{pair}}$	5
4	Structural Analysis: Why Extrapolation Fails	5
4.1	Empirical fits are degenerate	5
4.2	The leading correction is approximately constant in $\log q$. . .	5
4.3	Corrected values are consistent with the admissible window .	5
5	The Central Open Direction: $n_1(q)/q$	6
6	Discussion	6
6.1	What O25 establishes	6
6.2	Relation to O24	7
6.3	What remains open	7
7	Conclusion	7

1 Position of the Problem

1.1 What O16–O24 established

The spectral admissibility sub-programme has established, through O16–O24 [4, 5, 6, 1, 9, 11, 10, 12, 13], that the physically relevant observable is the canonical pair-level quantity

$$\sigma_{\text{pair}}^{\text{can}}(n) = \sigma_c(n) \sigma_{q-c}(n), \quad (1)$$

with capacity exponent $\delta_{\text{pair}} \approx 7.44$ consistent with the phenomenological target $\delta_{\text{pair}} \in [7.4, 10.6]$. The transfer chain $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$ was established unconditionally with respect to the fibre structure of Π in O24 [13].

The remaining open direction identified in O24 Section 5.1 is the extended numerical campaign at large primes across multiple conjugate pairs. O16 computed δ_{pair} for a single conjugate pair per prime (those pairs that

happened to appear simultaneously in the O12 random block sample [3]). No systematic coverage of all pairs was available.

1.2 Scope of O25

The present paper provides the first systematic computation of δ_{pair} across all $(q-1)/2$ conjugate pairs for each prime, with $M = 50$ independent block samples per pair. This yields:

- the full distribution of δ_{pair} across pairs at each prime,
- the mean $\bar{\delta}_{\text{pair}}(q)$ and inter-pair standard deviation σ_q ,
- the q -dependence of these quantities over $q \in \{29, 61, 101, 151\}$.

The paper does *not* claim to identify the asymptotic value δ_∞ . Instead, it establishes why direct extrapolation in q is structurally unreliable and identifies the correct analytical target: the ratio $n_1(q)/q$.

2 Setup and Protocol

2.1 Observable and block parameterisation

We follow the O12/O13 protocol [3, 2] without modification. For a prime q and a generic block $c_{\text{block}} = (c_1, c_2, c_3)$ with $c_i \neq 0$ and $c_1 + c_2 + c_3 \not\equiv 0 \pmod{q}$, the observable is

$$\sigma_c(n) = \frac{\delta r_n}{|S_n|}, \quad (2)$$

where δr_n is the number of Weil fingerprint vectors in BFS shell S_n that are linearly independent of the Gram–Schmidt span built over shells $0, \dots, n-1$.

2.2 Pair protocol

For each conjugate pair $(c, q-c)$ with $c \in \{1, \dots, (q-1)/2\}$:

1. Draw $M = 50$ independent pairs of blocks: $(c_1=c, c_2, c_3)$ and $(c'_1=q-c, c'_2, c'_3)$ with c_2, c_3, c'_2, c'_3 sampled uniformly and generically. This reproduces the O16 protocol of independently sampled blocks [4].
2. Compute $\sigma_c(n)$ and $\sigma_{q-c}(n)$ independently.
3. Set $\sigma_{\text{pair}}^{\text{can}}(n) = \sigma_c(n) \sigma_{q-c}(n)$.
4. Extract δ_{pair} by OLS regression of $\log \sigma_{\text{pair}}^{\text{can}}(n)$ vs $\log(n+1)$ over the fitting window $[n_0, n_1]$ (O16 convention).

2.3 Fitting window and BFS parameters

All runs use `bfs_frac` = 0.99 (full graph exploration) and auto-calibrated fitting windows from the actual `sigma_bar` after BFS, via the `find_fitting_window` procedure of O12 [3].

q	$n_{\max}^{\text{block}}$	shells	$ G_q^{\text{partial}} $	window $[n_0, n_1]$	n_1/q
29	8	28	24 145	[2, 4]	0.138
61	10	58	224 711	[2, 8]	0.131
101	20	95	1 019 997	[3, 11]	0.109
151	37	141	3 408 521	[3, 13]	0.086

Table 1: BFS and fitting parameters for each prime. The ratio n_1/q decreases with q , indicating that the fitting window has not yet entered the asymptotic regime $n_1 \sim \alpha q$.

3 Numerical Results

3.1 Main results table

q	pairs	M	$\bar{\delta}_{\text{pair}}$	σ_q	$\sigma_q/\bar{\delta}_{\text{pair}}$ (%)	valid	R_{global}^2
29	14	50	8.893	0.539	6.1	14/14	0.9998
61	30	50	9.983	0.259	2.6	30/30	0.9984
101	50	50	9.548	0.161	1.7	50/50	0.9975
151	75	50	9.125	0.139	1.5	75/75	0.9982

Table 2: Summary of O25 results. $\bar{\delta}_{\text{pair}}$: mean of δ_{pair} over all pairs. σ_q : inter-pair standard deviation. $\sigma_q/\bar{\delta}_{\text{pair}}$: coefficient of variation. “valid”: pairs for which the fitting window contained at least 2 non-zero data points.

3.2 Inter-pair concentration

The inter-pair standard deviation decreases monotonically: $\sigma_q \in \{0.539, 0.259, 0.161, 0.139\}$ for $q \in \{29, 61, 101, 151\}$. The coefficient of variation drops from 6.1% to 1.5%. This concentration confirms that δ_{pair} is a structural invariant of the Weil representation on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, as predicted by the verticality theorem of O24 [13]: the fibre structure of Π cannot perturb the observable rank, so inter-pair variance is purely a sampling effect that vanishes as $M \rightarrow \infty$.

3.3 q -dependence of $\bar{\delta}_{\text{pair}}$

The sequence $\bar{\delta}_{\text{pair}}(q)$ is not monotone: it rises from $q = 29$ to $q = 61$ ($8.89 \rightarrow 9.98$), then decreases from $q = 61$ to $q = 151$ ($9.98 \rightarrow 9.55 \rightarrow 9.13$). The non-monotone step at $q = 29$ is consistent with the very short fitting window ($[2,4]$, three points), which produces a less reliable estimate. The physically relevant trend is the monotone decrease from $q = 61$ onward, with apparent stabilisation between $q = 101$ and $q = 151$.

4 Structural Analysis: Why Extrapolation Fails

4.1 Empirical fits are degenerate

The sequence $\bar{\delta}_{\text{pair}}(q)$ for $q \in \{61, 101, 151\}$ is well described by multiple functional forms:

$$\bar{\delta}_{\text{pair}}(q) = \delta_{\infty} + \frac{a}{\log q}, \quad \bar{\delta}_{\text{pair}}(q) = \delta_{\infty} + \frac{a}{(\log q)^2}, \quad \bar{\delta}_{\text{pair}}(q) = \delta_{\infty} + \frac{a}{q^{\alpha}}. \quad (3)$$

All three fit the three data points with residuals below 0.06, while producing very different asymptotic values: $\delta_{\infty} \in \{5.32, 7.45, 5.80\}$ respectively.

This degeneracy is not a statistical artefact. It reflects a structural property of the problem: the leading finite-size correction to $\bar{\delta}_{\text{pair}}(q)$ is approximately constant in $\log q$, making the deviations from δ_{∞} indistinguishable from multiple slowly varying functions over the accessible range.

4.2 The leading correction is approximately constant in $\log q$

The observable $\sigma_c(n)$ is normalised by $|S_n|$, whose growth in the pre-saturation window $[n_0, n_1]$ is not yet in the asymptotic Bass–Guivarc’h regime $|S_n| \sim n^3$. O14 [7] identifies the dominant finite-size correction as

$$\delta_{\text{corr}}(q) = \delta_{\text{pair}}(q) - \eta \frac{\log q}{\log n_1(q)}, \quad (4)$$

with $\eta = 1/2$ derived from the Weil amplitude scaling. By the Window-Depth Theorem (O9 [8]), $n_1(q) = \Theta(q)$, so $\log n_1(q) \sim \log q$ and the correction factor $\log q / \log n_1(q) \approx 1$ to leading order. This means the dominant correction is *approximately constant*, not a power of $1/\log q$. Any fit of the form $\delta_{\infty} + f(q)$ for a slowly varying f will therefore describe the data well without carrying asymptotic meaning.

4.3 Corrected values are consistent with the admissible window

Applying (4) with $\eta = 1/2$ yields:

q	$\bar{\delta}_{\text{pair}}$	$\eta \log q / \log n_1$	δ_{corr}
61	9.983	0.988	8.995
101	9.548	0.962	8.586
151	9.125	0.978	8.147

Table 3: O14 normalization correction applied to O25 data. The corrected values $\delta_{\text{corr}}(q) \in [7.7, 9.0]$ fall within the admissible window $[7.4, 10.6]$ for all three primes. The correction factor $\eta \log q / \log n_1 \approx 0.97\text{--}0.99$ is approximately constant, as predicted by the $n_1 = \Theta(q)$ scaling.

The corrected values are consistent with an asymptotic value $\delta_\infty \in [7.4, 10.6]$. We emphasise that $\eta = 1/2$ is derived in O14 under assumptions that have not been verified in the exact O25 setting; the correction is therefore a *structural hypothesis* rather than a derived result. The observation that it brings all three data points into the admissible window is consistent with this hypothesis but does not constitute a proof.

5 The Central Open Direction: $n_1(q)/q$

The analysis of Section 4 identifies the ratio $n_1(q)/q$ as the key quantity controlling convergence. From Table 1: $n_1(q)/q \in \{0.138, 0.131, 0.109, 0.086\}$ for $q \in \{29, 61, 101, 151\}$. The ratio is *decreasing*, not yet constant. This means the system is not in the asymptotic regime $n_1 \sim \alpha q$ for any fixed constant α , and the correction of Section 4.2 is itself q -dependent.

Two sub-questions are open:

1. Does $n_1(q)/q$ stabilise as $q \rightarrow \infty$, and if so, to what value α ?
2. If $n_1(q) = \alpha q + O(q^\beta)$ with $\beta < 1$, what is the resulting correction to $\bar{\delta}_{\text{pair}}(q)$ at next-to-leading order?

Answering these questions would allow an analytical prediction of δ_∞ from the structure of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, independently of numerical extrapolation.

6 Discussion

6.1 What O25 establishes

Three results are established:

1. **Numerical robustness.** $\delta_{\text{pair}}(q)$ converges and the inter-pair variance decreases strongly with q . At $q = 151$, the coefficient of variation is 1.5% over 75 pairs with $M = 50$ samples each. This confirms the structural prediction of O24 [13].

2. **Structural non-identifiability of δ_∞ by extrapolation.** The leading finite-size correction is approximately constant in $\log q$, making all empirical fits degenerate. Direct extrapolation in q cannot identify the asymptotic value.
3. **Consistency with the admissible window under O14 correction.** When the normalization correction of O14 is applied, all three data points at $q \in \{61, 101, 151\}$ fall within $[7.4, 10.6]$. This is consistent with the interpretation that the apparent decrease of raw $\bar{\delta}_{\text{pair}}(q)$ is a finite-size normalization effect.

6.2 Relation to O24

O24 [13] established the structural integrity of the chain $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$ unconditionally. O25 provides the quantitative numerical content: it shows that the values of δ_{pair} measured at accessible primes are consistent with the admissible window, once the finite-size normalization structure is accounted for analytically. The two papers together close the numerical open direction of the O-series at the level of structural consistency.

6.3 What remains open

The following directions are explicitly open:

- Analytical derivation of the asymptotic ratio $n_1(q)/q$ as $q \rightarrow \infty$.
- Verification of the O14 hypothesis $\eta = 1/2$ in the exact O25 pipeline setting.
- Extension of the campaign to $q = 211$ and beyond, to constrain the rate of convergence of $n_1(q)/q$.

7 Conclusion

The convergence of $\delta_{\text{pair}}(q)$ is robust and the inter-pair concentration is strong, confirming that δ_{pair} is a structural invariant of the admissibility programme. Empirical fits in q provide excellent descriptions of the data but carry no asymptotic meaning, because the leading finite-size correction is governed by the BFS window depth $n_1(q)$ and is approximately constant in $\log q$. When the O14 normalization correction is applied, the corrected values $\delta_{\text{corr}}(q) \in [7.7, 9.0]$ are consistent with an asymptotic value in the admissible window $[7.4, 10.6]$, for all tested primes $q \in \{61, 101, 151\}$. The identification of δ_∞ therefore requires analytical control of the ratio $n_1(q)/q$, which is identified as the central open direction for subsequent work.

Data and code availability

The numerical pipeline `o25_paired_pipeline.py`, the checkpoint files `q{q}_o25.npz` for $q \in \{29, 61, 101, 151\}$, and the analysis script `o25_analysis.py` are available at Zenodo [DOI to be assigned upon deposit].

Acknowledgements

References

- [1] Jérôme Beau. About canonical pair observables in weil blocks: Structure of the residual amplitude factor and pipeline-independent formulation, 2026. Spectral admissibility sub-program paper.
- [2] Jérôme Beau. Asymptotic stability of exact weil-block capacity on heisenberg graphs: Extended prime range, variance reduction, and requalification of the $\delta \rightarrow \beta^*$ tension, 2026. Spectral admissibility sub-program paper.
- [3] Jérôme Beau. Exact weil-block projective capacity on heisenberg graphs: Resolving the final obstruction to δ extraction, 2026. Spectral admissibility sub-program paper.
- [4] Jérôme Beau. Fibre admissibility and resolution of the capacity exponent discrepancy, 2026. Spectral admissibility sub-program paper.
- [5] Jérôme Beau. Fibre-level admissibility from conjugate weil blocks: Structural derivation of the pair observable, 2026. O17 paper of Cosmochrony Spectral Admissibility sub-Program.
- [6] Jérôme Beau. Minimal fibre structure of the non-injective projection from born–infeld indiscernability: Derivation of the parity involution, 2026. Spectral admissibility sub-program paper.
- [7] Jérôme Beau. Observable-class mismatch and the corrected $\delta \mapsto \beta^*$ relation on heisenberg graphs: Block normalisation, central-phase contribution, 2026. O14 spectral admissibility subprogram paper.
- [8] Jérôme Beau. Projective capacity beyond expanders: Polynomial-growth relaxation graphs and the capacity exponent, 2026. Spectral admissibility sub-program paper.
- [9] Jérôme Beau. Projective persistence and the physical sub-spectrum: A dynamical selection criterion for the capacity exponent, 2026. Spectral admissibility sub-program paper.

- [10] Jérôme Beau. Shell-alignment from projection locking: A discrete admissibility theorem under born-infeld saturation, 2026. Spectral admissibility sub-program paper.
- [11] Jérôme Beau. Shell-level saturation and the fibre-level transfer $\delta_{\text{pair}} \rightarrow \beta^*$, 2026. Spectral admissibility sub-program paper.
- [12] Jérôme Beau. Three stable directions from quaternionic minimality: Derivation of $\sigma_c(n_3) = 3$ from born-infeld fibre admissibility, 2026. Spectral admissibility sub-program paper.
- [13] Jérôme Beau. Vertical non-injectivity and the stability of the observable rank: Closing the unconditional transfer $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$, 2026. O24 spectral admissibility subprogram paper.